

FagusED · A-Level 9231
1.1 Roots of polynomial equations

What is the central idea in Roots of polynomial equations?

9231-01-C1

QUESTION

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1.1 Roots of polynomial equations

Which mechanism, condition or distinction matters in Roots of polynomial equations?

9231-01-C2

QUESTION

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1.1 Roots of polynomial equations

A monic cubic has roots 1, 2 and 4. Write equation whose roots are their reciprocals.

9231-01-P

QUESTION

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1.2 Rational functions and graphs

What is the central idea in Rational functions and graphs?

9231-02-C1

QUESTION

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1.2 Rational functions and graphs

Which mechanism, condition or distinction matters in Rational functions and graphs?

9231-02-C2

QUESTION

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1.2 Rational functions and graphs

Find vertical and horizontal asymptotes of $y=(2x+1)/(x-3)$ and identify the x-intercept.

9231-02-P

QUESTION

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1.3 Summation of series

What is the central idea in Summation of series?

9231-03-C1

QUESTION

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1.3 Summation of series

Which mechanism, condition or distinction matters in Summation of series?

9231-03-C2

QUESTION

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1.3 Summation of series

Find Σ from $r=1$ to n of $1/[r(r+1)]$.

9231-03-P

QUESTION

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1.4 Matrices

What is the central idea in Matrices?

9231-04-C1

QUESTION

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1.4 Matrices

Which mechanism, condition or distinction matters in Matrices?

9231-04-C2

QUESTION

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1.4 Matrices

For $A=[[2,0],[0,-1]]$, find eigenvalues and determinant, and describe transformation of x and y coordinates.

9231-04-P

QUESTION

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1.5 Polar coordinates

What is the central idea in Polar coordinates?

9231-05-C1

QUESTION

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1.5 Polar coordinates

Which mechanism, condition or distinction matters in Polar coordinates?

9231-05-C2

QUESTION

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1.5 Polar coordinates

Find area swept by polar curve $r=2$ from $\theta=0$ to $\theta=\pi/2$.

9231-05-P

QUESTION

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1.1 Roots of polynomial equations

Reciprocal roots 1, $\frac{1}{2}$, $\frac{1}{4}$;
multiply $(x-1)(2x-1)(4x-1)$; $8x^3 - 14x^2 + 7x - 1 = 0$.

9231-01-P

ANSWER

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1.1 Roots of polynomial equations

Transformed roots can be found without solving the original polynomial by substituting a new variable and clearing denominators carefully.

9231-01-C2

ANSWER

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1.1 Roots of polynomial equations

For polynomial equations, Vieta's formulas link symmetric sums/products of roots to coefficients. If coefficients are real, non-real complex roots occur in conjugate pairs.

9231-01-C1

ANSWER

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1.2 Rational functions and graphs

Denominator zero gives $x=3$;
ratio leading coefficients $y=2$;
numerator zero gives $x=-\frac{1}{2}$.

9231-02-P

ANSWER

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1.2 Rational functions and graphs

Differentiation locates stationary points; sign charts show intervals where the curve rises and falls. A cancelled factor may produce a hole rather than a vertical asymptote.

9231-02-C2

ANSWER

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1.2 Rational functions and graphs

A rational graph is controlled by zeros of numerator, excluded denominator values, asymptotes and possible cancellations. Divide polynomials for an oblique asymptote.

9231-02-C1

ANSWER

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1.3 Summation of series

Partial fractions $\frac{1}{r(r+1)} = \frac{1}{r} - \frac{1}{r+1}$; cancellation; $\text{sum} = 1 - \frac{1}{n+1} = \frac{n}{n+1}$.

9231-03-P

ANSWER

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1.3 Summation of series

Induction is useful to prove a proposed result: verify base case, assume $n=k$, then derive $n=k+1$ explicitly. An infinite geometric sum requires $\text{abs}(r) < 1$; never apply it to divergent series.

9231-03-C2

ANSWER

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1.3 Summation of series

Use finite-sum identities for r , r^2 , r^3 and partial fractions for telescoping sums.

9231-03-C1

ANSWER

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1.4 Matrices

Eigenvalues 2 and -1 ; $\det A = -2$; x doubles while y is reflected across x -axis.

9231-04-P

ANSWER

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1.4 Matrices

Eigenvectors retain direction under transformation, with eigenvalue scale factor. An inverse exists only when determinant is nonzero.

9231-04-C2

ANSWER

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1.4 Matrices

A matrix represents a linear transformation. Matrix multiplication composes transformations in reverse order of application; determinants reveal area scale and invertibility.

9231-04-C1

ANSWER

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1.5 Polar coordinates

$A = \frac{1}{2} \int_0^{\pi/2} r^2 d\theta = \frac{1}{2} \times 4 \times \pi/2 = \pi$ square units.

9231-05-P

ANSWER

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1.5 Polar coordinates

For a polar curve, area between $\theta=a$ and b is $\frac{1}{2} \int r^2 d\theta$; intersections may require considering repeated representations of the same point.

9231-05-C2

ANSWER

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1.5 Polar coordinates

Polar coordinates use $x=r \cos \theta$, $y=r \sin \theta$. Negative r represents a point in the opposite direction to θ , so specify conventions carefully.

9231-05-C1

ANSWER

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1.6 Vectors

What is the central idea in Vectors?

9231-06-C1

QUESTION

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1.6 Vectors

Which mechanism, condition or distinction matters in Vectors?

9231-06-C2

QUESTION

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1.6 Vectors

Find angle between $(1,0,0)$ and $(1,1,0)$, giving exact value.

9231-06-P

QUESTION

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1.7 Proof by induction

What is the central idea in Proof by induction?

9231-07-C1

QUESTION

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1.7 Proof by induction

Which mechanism, condition or distinction matters in Proof by induction?

9231-07-C2

QUESTION

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1.7 Proof by induction

Prove $1+3+\dots+(2n-1)=n^2$ by induction.

9231-07-P

QUESTION

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2.1 Hyperbolic functions

What is the central idea in Hyperbolic functions?

9231-08-C1

QUESTION

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2.1 Hyperbolic functions

Which mechanism, condition or distinction matters in Hyperbolic functions?

9231-08-C2

QUESTION

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2.1 Hyperbolic functions

Show $\cosh^2 x - \sinh^2 x = 1$ directly from exponential definitions.

9231-08-P

QUESTION

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2.2 Matrices

What is the central idea in Matrices?

9231-09-C1

QUESTION

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2.2 Matrices

Which mechanism, condition or distinction matters in Matrices?

9231-09-C2

QUESTION

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2.2 Matrices

For $A = \begin{bmatrix} 3 & 0 \\ 0 & 5 \end{bmatrix}$, give eigenvalues and one eigenvector for each.

9231-09-P

QUESTION

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2.3 Differentiation

What is the central idea in Differentiation?

9231-10-C1

QUESTION

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2.3 Differentiation

Which mechanism, condition or distinction matters in Differentiation?

9231-10-C2

QUESTION

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2.3 Differentiation

Find dy/dx on $x^2 + y^2 = 25$ at $(3,4)$ and state tangent gradient.

9231-10-P

QUESTION

Dot product = 1; lengths 1 and $\sqrt{2}$; $\cos \theta = 1/\sqrt{2}$, $\theta = \pi/4$.

9231-06-P

ANSWER

Dot product tests perpendicularity and angles; cross product gives a perpendicular vector and parallelogram area.

9231-06-C2

ANSWER

A vector line can be $r = a + \lambda b$. To find intersections, equate components and solve for distinct parameters. A plane $r \cdot n = d$ has normal vector n .

9231-06-C1

ANSWER

$n=1$: $1=1^2$. Assume sum through k is k^2 ; add $2(k+1)-1=2k+1$ to get $k^2+2k+1=(k+1)^2$.

9231-07-P

ANSWER

The inductive hypothesis is used at a visible algebraic step, not merely asserted. For divisibility, rewrite the $n=k+1$ expression as a known divisible term plus a clearly divisible remainder.

9231-07-C2

ANSWER

Induction proves a proposition for all positive integers by establishing a base case and showing $P(k)$ implies $P(k+1)$.

9231-07-C1

ANSWER

Difference of squares:
 $((e^x + e^{-x})^2 - (e^x - e^{-x})^2)/4 = 4e^x e^{-x}/4 = 1$.

9231-08-P

ANSWER

Inverse hyperbolic functions can be written using logarithms by solving a quadratic in e^x .

9231-08-C2

ANSWER

$\sinh x = (e^x - e^{-x})/2$ and $\cosh x = (e^x + e^{-x})/2$. Their identity $\cosh^2 x - \sinh^2 x = 1$ contrasts with trigonometric identities; derivatives are $\cosh$ and $\sinh$ respectively.

9231-08-C1

ANSWER

Eigenvalues 3 and 5; $v_3 = (1, 0)$, $v_5 = (0, 1)$; nonzero scalar multiples accepted.

9231-09-P

ANSWER

Solve $\det(A - \lambda I) = 0$ for eigenvalues; an eigenvector satisfies $(A - \lambda I)v = 0$. Repeated eigenvalues need enough independent eigenvectors for diagonalisation.

9231-09-C2

ANSWER

A Level matrix work extends to three-dimensional transformations, eigenvalues and diagonalisation where applicable.

9231-09-C1

ANSWER

Differentiate: $2x + 2y$ $y' = 0$; $y' = -x/y$; at $(3, 4)$, $-3/4$.

9231-10-P

ANSWER

For polar curves, compute derivatives through $x = r \cos \theta$ and $y = r \sin \theta$. Higher derivatives locate turning-point type and curvature.

9231-10-C2

ANSWER

Implicit differentiation treats y as a function of x ; derivatives of y terms acquire dy/dx . Parametric differentiation uses $dy/dx = (dy/dt)/(dx/dt)$, provided $dx/dt \neq 0$.

9231-10-C1

ANSWER

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2.4 Integration

What is the central idea in Integration?

9231-11-C1

QUESTION

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2.4 Integration

Which mechanism, condition or distinction matters in Integration?

9231-11-C2

QUESTION

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2.4 Integration

Evaluate $\int$ from 0 to 1 of $x e^x dx$.

9231-11-P

QUESTION

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2.5 Complex numbers

What is the central idea in Complex numbers?

9231-12-C1

QUESTION

FagusED · A-Level 9231
2.5 Complex numbers

Which mechanism, condition or distinction matters in Complex numbers?

9231-12-C2

QUESTION

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2.5 Complex numbers

Find the two square roots of -4 in complex numbers and give their moduli.

9231-12-P

QUESTION

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2.6 Differential equations

What is the central idea in Differential equations?

9231-13-C1

QUESTION

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2.6 Differential equations

Which mechanism, condition or distinction matters in Differential equations?

9231-13-C2

QUESTION

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2.6 Differential equations

Solve $dy/dx=3y$ given $y(0)=2$.

9231-13-P

QUESTION

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3.1 Motion of a projectile

What is the central idea in Motion of a projectile?

9231-14-C1

QUESTION

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3.1 Motion of a projectile

Which mechanism, condition or distinction matters in Motion of a projectile?

9231-14-C2

QUESTION

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3.1 Motion of a projectile

A particle is projected at 25 m s^{-1} at 30° above horizontal. Find maximum height, $g=10 \text{ m s}^{-2}$.

9231-14-P

QUESTION

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3.2 Equilibrium of a rigid body

What is the central idea in Equilibrium of a rigid body?

9231-15-C1

QUESTION

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3.2 Equilibrium of a rigid body

Which mechanism, condition or distinction matters in Equilibrium of a rigid body?

9231-15-C2

QUESTION

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3.2 Equilibrium of a rigid body

A horizontal uniform beam 4 m long has weight 80 N at centre; a support at one end and upward force at other hold it. Find upward force at far end.

9231-15-P

QUESTION

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2.4 Integration

Parts gives antiderivative $e^x(x-1)$; evaluate at 1 minus at 0: $0 - (-1) = 1$.

9231-11-P

ANSWER

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2.4 Integration

Definite integrals can represent signed area, so use absolute value or split where actual geometric area is asked.

9231-11-C2

ANSWER

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2.4 Integration

Integration may require substitution, partial fractions, integration by parts or reduction formulae. Check domains for logarithmic antiderivatives; $\int 1/x \, dx = \ln|x| + C$.

9231-11-C1

ANSWER

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2.5 Complex numbers

$z^2 = -4$ gives $z = \pm 2i$; each modulus is 2.

9231-12-P

ANSWER

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2.5 Complex numbers

Conjugation reflects across real axis. Roots of a complex number lie evenly spaced around a circle; De Moivre's theorem gives powers.

9231-12-C2

ANSWER

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2.5 Complex numbers

Write $z = x + iy$, modulus $\sqrt{x^2 + y^2}$, argument with quadrant check; polar form $r(\cos \theta + i \sin \theta)$ simplifies multiplication and powers.

9231-12-C1

ANSWER

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2.6 Differential equations

Separate $dy/y = 3dx$;
 $\ln|x| = 3x + C$; $y = 2e^{3x}$.

9231-13-P

ANSWER

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2.6 Differential equations

Particular solutions require applying the initial condition after finding the general solution; an omitted constant loses a family of solutions.

9231-13-C2

ANSWER

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2.6 Differential equations

Separate variables only when dy/dx can be arranged as $f(y)g(x)$. For linear first-order equations use an integrating factor when required.

9231-13-C1

ANSWER

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3.1 Motion of a projectile

$u_y = 25 \sin 30^\circ = 12.5$; at top
 $v_y = 0$, $h = u_y^2 / (2g) = 7.8125$ m.

9231-14-P

ANSWER

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3.1 Motion of a projectile

Use common time to combine components; at maximum height vertical velocity is zero. Range at equal launch/landing height is $u^2 \sin(2\theta) / g$.

9231-14-C2

ANSWER

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3.1 Motion of a projectile

Resolve launch velocity into perpendicular components. Horizontal acceleration is zero and vertical acceleration is $-g$ when air resistance is ignored.

9231-14-C1

ANSWER

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3.2 Equilibrium of a rigid body

Take moments about support:
 $F \times 4 = 80 \times 2$; $F = 40$ N.

9231-15-P

ANSWER

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3.2 Equilibrium of a rigid body

Take moments about a point that removes unknown forces to solve efficiently, then check force components and physical sign.

9231-15-C2

ANSWER

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3.2 Equilibrium of a rigid body

Rigid-body equilibrium requires both resultant force zero and resultant moment zero. Weight acts through centre of mass; contact reactions and friction depend on geometry.

9231-15-C1

ANSWER

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3.3 Circular motion

What is the central idea in Circular motion?

9231-16-C1

QUESTION

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3.3 Circular motion

Which mechanism, condition or distinction matters in Circular motion?

9231-16-C2

QUESTION

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3.3 Circular motion

A 0.5 kg mass travels at 4 m s^{-1} around radius 2 m on a horizontal frictionless table. Determine string tension.

9231-16-P

QUESTION

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3.4 Hooke's law

What is the central idea in Hooke's law?

9231-17-C1

QUESTION

FagusED · A-Level 9231
3.4 Hooke's law

Which mechanism, condition or distinction matters in Hooke's law?

9231-17-C2

QUESTION

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3.4 Hooke's law

A spring of stiffness 200 N m^{-1} stretches 0.10 m. Calculate tension and energy stored.

9231-17-P

QUESTION

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3.5 Linear motion under a variable force

What is the central idea in Linear motion under a variable force?

9231-18-C1

QUESTION

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3.5 Linear motion under a variable force

Which mechanism, condition or distinction matters in Linear motion under a variable force?

9231-18-C2

QUESTION

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3.5 Linear motion under a variable force

A 2 kg particle at rest is pushed by $F=6x \text{ N}$ from $x=0$ to $x=2 \text{ m}$. Find speed at $x=2 \text{ m}$.

9231-18-P

QUESTION

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3.6 Momentum

What is the central idea in Momentum?

9231-19-C1

QUESTION

FagusED · A-Level 9231
3.6 Momentum

Which mechanism, condition or distinction matters in Momentum?

9231-19-C2

QUESTION

FagusED · A-Level 9231
3.6 Momentum

A 1 kg trolley at 4 m s^{-1} hits a stationary 1 kg trolley; they stick. Find common velocity and kinetic-energy change.

9231-19-P

QUESTION

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4.1 Continuous random variables

What is the central idea in Continuous random variables?

9231-20-C1

QUESTION

FagusED · A-Level 9231
4.1 Continuous random variables

Which mechanism, condition or distinction matters in Continuous random variables?

9231-20-C2

QUESTION

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4.1 Continuous random variables

For $f(x)=kx$ on $0 \leq x \leq 2$, zero elsewhere, find k and $P(X>1)$.

9231-20-P

QUESTION

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3.3 Circular motion

Radial acceleration $v^2/r=16/2=8$ m s^{-2} ; tension supplies resultant; $T=ma=4$ N.

9231-16-P

ANSWER

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3.3 Circular motion

State the chosen positive radial direction before writing Newton's second law.

9231-16-C2

ANSWER

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3.3 Circular motion

For circular movement, acceleration points inward with magnitude $v^2/r = \omega^2 r$. In vertical circles, radial resultant changes with position because the radial component of weight changes.

9231-16-C1

ANSWER

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3.4 Hooke's law

$T=kx=20$ N; energy $=\frac{1}{2}kx^2 = 1.0$ J.

9231-17-P

ANSWER

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3.4 Hooke's law

For systems with two springs, identify whether common force (series) or common extension (parallel) applies before replacing them by an effective stiffness.

9231-17-C2

ANSWER

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3.4 Hooke's law

A Hookean spring obeys $T=kx$ for extension x within elastic limit. Elastic energy stored is $\frac{1}{2}kx^2$, the area under its force-extension graph.

9231-17-C1

ANSWER

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3.5 Linear motion under a variable force

Work $\int_0^2 6x \, dx = 12$ J; $\frac{1}{2}(2)v^2 = 12$, $v = \sqrt{12} \approx 3.46$ m s^{-1} .

9231-18-P

ANSWER

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3.5 Linear motion under a variable force

The work-energy method $\int F \, dx = \Delta(\frac{1}{2}mv^2)$ can simplify finding speed without time.

9231-18-C2

ANSWER

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3.5 Linear motion under a variable force

For variable force, Newton's law is $m \, dv/dt = F(t, x, v)$. If acceleration is a function of position, use $a = v \, dv/dx$, then integrate.

9231-18-C1

ANSWER

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3.6 Momentum

Momentum gives $(1)(4) = (2)v$, so $v = 2$ m s^{-1} ; initial KE 8 J, final 4 J, change -4 J.

9231-19-P

ANSWER

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3.6 Momentum

The coefficient of restitution e compares relative separation speed to approach speed along the line of impact, ranging from 0 to 1 in usual collision models.

9231-19-C2

ANSWER

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3.6 Momentum

Impulse is $\int F dt = \text{change in momentum}$. Momentum is conserved for an isolated system; kinetic energy is conserved only in elastic collisions.

9231-19-C1

ANSWER

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4.1 Continuous random variables

$1 = \int_0^2 kx \, dx = 2k$, so $k = 1/2$;
 $P(X > 1) = \int_1^2 x/2 \, dx = 3/4$.

9231-20-P

ANSWER

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4.1 Continuous random variables

Find $E(X) = \int xf(x) \, dx$ and $\text{Var}(X) = E(X^2) - E(X)^2$; transform a distribution by checking domains and Jacobian when needed.

9231-20-C2

ANSWER

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4.1 Continuous random variables

A probability density integrates to one and probabilities are areas, not heights. For continuous X , $P(X=a)=0$.

9231-20-C1

ANSWER

What is the central idea in Inference using normal and t-distributions?

9231-21-C1

QUESTION

Which mechanism, condition or distinction matters in Inference using normal and t-distributions?

9231-21-C2

QUESTION

For $n=16$, sample $s=4$ and mean $=50$, calculate standard error and degrees of freedom for a t interval.

9231-21-P

QUESTION

What is the central idea in Chi-square tests?

9231-22-C1

QUESTION

Which mechanism, condition or distinction matters in Chi-square tests?

9231-22-C2

QUESTION

Observed counts 30 and 20; expected 25 and 25. Calculate χ^2 and degrees of freedom for two categories with no fitted parameters.

9231-22-P

QUESTION

What is the central idea in Non-parametric tests?

9231-23-C1

QUESTION

Which mechanism, condition or distinction matters in Non-parametric tests?

9231-23-C2

QUESTION

In six paired observations, five differences are positive and one negative. Give two-sided sign-test p-value under H_0 .

9231-23-P

QUESTION

What is the central idea in Probability generating functions?

9231-24-C1

QUESTION

Which mechanism, condition or distinction matters in Probability generating functions?

9231-24-C2

QUESTION

A variable has $P(X=0)=0.3$ and $P(X=1)=0.7$. Find PGF and $E(X)$ using derivative.

9231-24-P

QUESTION

$s/\sqrt{n}=4/4=1$; degrees of freedom 15.

9231-21-P

ANSWER

For known σ and appropriate normal conditions use z ; with unknown σ and small normal sample use Student's t with $n-1$ degrees of freedom. State null/alternative, assumptions, test statistic and conclusion in context.

9231-21-C2

ANSWER

A confidence interval estimates a population parameter with stated coverage procedure.

9231-21-C1

ANSWER

$\chi^2=(30-25)^2/25+(20-25)^2/25=2.0$; $df=2-1=1$.

9231-22-P

ANSWER

Association tests use expected count row total $\times$ column total/grand total.

9231-22-C2

ANSWER

A chi-square goodness-of-fit statistic sums $(O-E)^2/E$ across categories. Expected counts must be large enough for approximation; pool categories where needed and adjust degrees of freedom for constraints/estimated parameters.

9231-22-C1

ANSWER

$P(X \geq 5) = C(6,5)/64 + C(6,6)/64 = 7/64$; two-sided $p = 14/64 = 0.21875$.

9231-23-P

ANSWER

Wilcoxon signed-rank additionally uses magnitudes via ranks, subject to its assumptions.

9231-23-C2

ANSWER

Rank-based tests use order rather than a specified normal distribution. A sign test checks direction of paired changes after discarding zero differences; under null, plus signs follow $\text{Binomial}(n, 0.5)$.

9231-23-C1

ANSWER

$G(t) = 0.3 + 0.7t$; $G'(t) = 0.7$;
 $E(X) = G'(1) = 0.7$.

9231-24-P

ANSWER

$G(1) = 1$, $G'(1) = E(X)$, and $G''(1) + G'(1) - G'(1)^2 = \text{Var}(X)$. Independent sums multiply PGFs, allowing convolution without listing every outcome.

9231-24-C2

ANSWER

A probability generating function $G_X(t) = E(t^X) = \sum_p t^p$ for non-negative integer X .

9231-24-C1

ANSWER